CS 162
Computer Architecture
Winter 2026

Distributed: Feb 11th, 2026

Due: 4:59pm Feb 19th, 2026

Points: 100

1. **(10 pts)** Fill the table below

TLB	Page Table	Cache	Possible? (yes or no)	Under what circumstances?
Hit	Hit	Hit		
Hit	Hit	Miss		
Miss	Hit	Hit		
Miss	Hit	Miss		
Miss	Miss	Miss		
Hit	Miss	Hit/ Miss		
Miss	Miss	Hit		

2. **(50 pts)** Assume the $D cache is initially filled with **word** address 0, 1, 2, …15 memory data (referenced by the datapath in that order). Below is the next sequence of data memory address references given as **word** addresses.
   1, 20, 2, 3, 4, 18, 5, 19, 33, 34, 1, 4

   a) Assuming a direct-mapped $D cache with one-word blocks and a total size of 16 blocks, list if each reference is a hit or a miss. Show the contents of the $D cache after the last reference. What is the hit rate for this reference string?

b) Now, assuming a direct-mapped $D cache with two-word blocks and a total size of 8 blocks, list if each reference is a hit or a miss. Show the state of the $D cache after the last reference. What is the hit rate for this reference string?

c) Now, assuming a set-associative $D cache with two ways, two-word blocks and a total size of 8 blocks, list if each reference is a hit or a miss (assume LRU replacement). Show the state of the $D cache after the last reference. What is the hit rate for this reference string?

3. **(20 pts)** For the following two sub-questions, the $I cache has a hit rate of 90%, a hit/access latency of 1 cycle, and a miss penalty of 100 cycles. The $D cache has a hit rate of 85%, a 2 cycle hit/access latency, a miss penalty of 120 cycles, and memory accesses (lw and sw) that are 30% of the instruction mix.

   a) What is the Average Memory Access Time? Note: The cache must be accessed after memory returns the data.

   b) Calculate the CPI taking into account stalls due to $D cache and $I cache misses. Report the CPI into additional CPI due to $I cache stalls, additional CPI due to $D cache stalls, and overall CPI. The base CPI using a "perfect" memory system is 1.0. Assume everything else is working perfectly - loads never stall a dependent instruction. Also assume the processor waits for stores to finish when they miss in the $D cache, and that $I cache misses and $D cache misses never occur at the same time.

4. **(20 pts)** Recall the cache optimizations we have discussed in class. Assuming that you are designing the memory system for the following two types of application domains, please choose the optimization strategies for cache and justify why you choose them. Note that you may choose multiple cache optimizations as long as you provide sound reasons.
   a) Graph applications dealing with irregular graphs (e.g., twitter user connection diagrams). **Hint:** random memory access pattern
   b) Streaming application dealing with regular images and videos. **Hint:** regular memory access pattern.